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NO.1 PVT. UNIVERSITY IN ACADEMIC REPUTATION IN INDIA

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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 05 – Lecture 03: Applications in Education, Insurance, Retail, Transport & Energy (Wrap-up)

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Topic focus: broad application map + deployment checklist + revision

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.

Quick Links

Domains Checklist Wrap-up Revision Summary

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Domains

Checklist

Wrap-up

Revision

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

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- Apply a simple "what could go wrong" checklist to an AI system.
- Answer key revision questions across Units I–V.

Abbreviations & Keywords

- KPI: key performance indicator

ETA: estimated time of arrival

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- Drift: patterns change over time
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- SLA: service-level agreement (latency/uptime constraints)
- Objective mismatch: wrong metric for the real goal

Education

- personalization / recommendations

Key concerns: fairness, privacy of student data, transparency.

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- automated feedback and assessment support

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- dropout risk prediction

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Insurance

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Retail & Supply Chain

- demand forecasting

Key concerns: drift (seasonality), feedback loops, data quality.

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- recommendation systems

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Retail & Supply Chain

- demand forecasting
- recommendation systems
- inventory optimization

Key concerns: drift (seasonality), feedback loops, data quality.

Transportation & Logistics

- route optimization

Key concerns: safety, real-time constraints, sensor reliability.

Transportation & Logistics

- route optimization
- ETA prediction

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Transportation & Logistics

- route optimization
- ETA prediction
- predictive maintenance

Key concerns: safety, real-time constraints, sensor reliability.

Energy & Utilities

- load forecasting

Key concerns: reliability, rare-event detection, monitoring.

Energy & Utilities

- load forecasting
- fault detection

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Energy & Utilities

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- fault detection
- demand-response optimization

Key concerns: reliability, rare-event detection, monitoring.

What Could Go Wrong? (Checklist)

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5. Security/privacy risks and misuse

Course Wrap-up (Big Picture)

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- Unit V: applications + responsible AI practices

Revision Questions (Sample)

1. What is entailment and how is it tested using refutation?
2. Why do we need train/validation/test splits?
3. Compare regression vs classification and their metrics.
4. What does PCA optimize and why standardize before PCA?
5. Give two examples of drift and how to monitor it.

Summary

- AI/ML applications are everywhere, but success depends on evaluation and constraints.
- Use a deployment checklist: metric, data, fairness, drift, security.
- Revision: connect logic + learning + applications.

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Exit Questions

1. Pick one domain and propose an ML task + suitable metric.
2. List three items from the "what could go wrong" checklist.
3. Why is explainability often required in insurance applications?
4. Give one example of a feedback loop in retail recommendations.